

Using Machine Learning to Predict Antimicrobial Resistance (AMR) Phenotypes

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Motivation

- Antimicrobial resistance (AMR) is a critical global public health concern due to the rise in hospital-acquired infections and heightened levels of illness and death.
- Researchers can utilize ML techniques to predict AMR phenotypes (resistant or susceptible)
- The objective of this study is to enhance AMR prediction using ML methods utilized in the cybersecurity domain due to the similarities between AMR and malware datasets.
- We also explore whether SMOTE augmentation improves model performance on low-sample AMR datasets.

Dataset & Features

- AMR metadata collection by ~BV-BRC (Bacterial and Viral Bioinformatics Center)
 - In genome name column, *Enterococcus faecium*
 - In antibiotic column, vancomycin
 - classify “resistant” and “susceptible”
- Feature Design
 - 562.48345.fna -> 562.48345.fna.txt

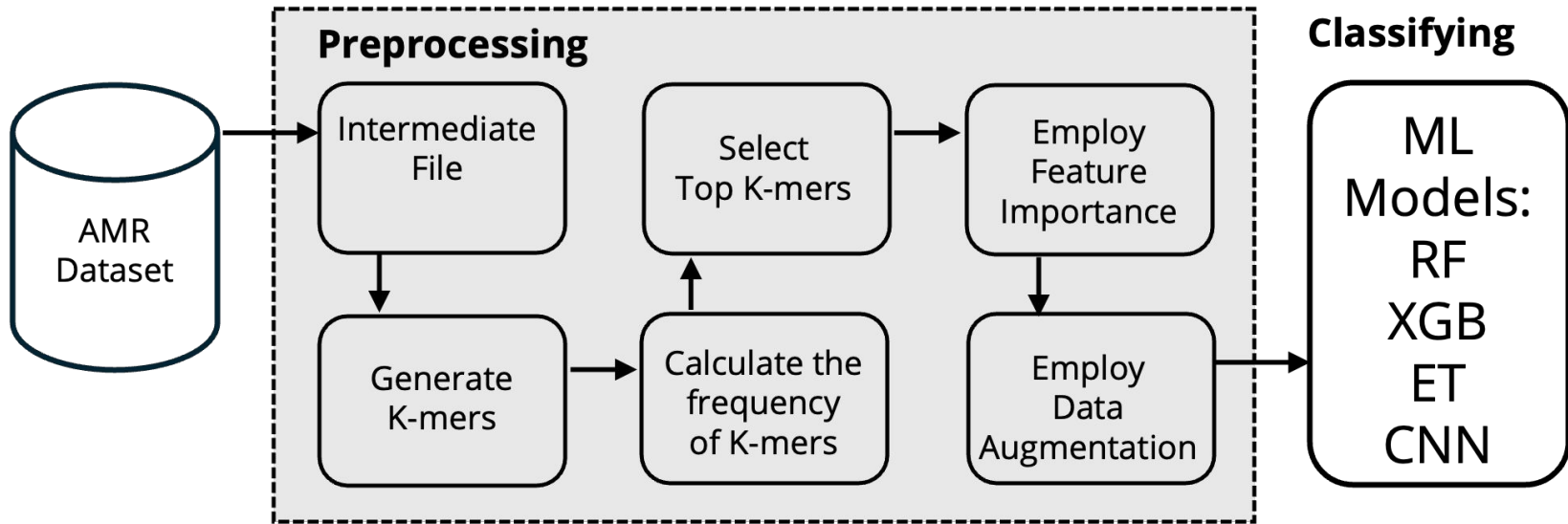
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cagcctcgat
agcctcgata
gcctcgatat
ctcggatata
ctcgatataa
tcgatataatc
gatataatcc
atataatcca
```

- Details of the dataset used for the experiment

Genome Name	Antibiotic	Resistant			Susceptible		
		Count	Avg	STD	Count	Avg	STD
E. faecium	vancomycin (VAN)	1,229	2.81 MB	0.99 MB	1,149	2.76 MB	0.84 MB
E. coli	ampicillin (AMP)	3,026	4.98 MB	0.37 MB	2,577	4.90 MB	0.25 MB
A. baumannii	amikacin (AMK)	727	3.83 MB	0.68 MB	370	3.93 MB	0.32 MB
E. coli	gentamicin (GEN)	797	4.78 MB	1.01 MB	6,132	4.90 MB	0.46 MB
E. coli	trimethoprim/sulfamethoxazole (SXT)	1,081	4.98 MB	0.30 MB	2,477	4.93 MB	0.23 MB
E. faecium	ampicillin (AMP)	531	2.97 MB	0.54 MB	209	2.81 MB	0.60 MB
S. enterica	ampicillin (AMP)	1,954	4.76 MB	0.35 MB	2,709	4.68 MB	0.30 MB
S. enterica	chloramphenicol (CHL)	1,083	4.78 MB	0.38 MB	3,486	4.64 MB	0.30 MB

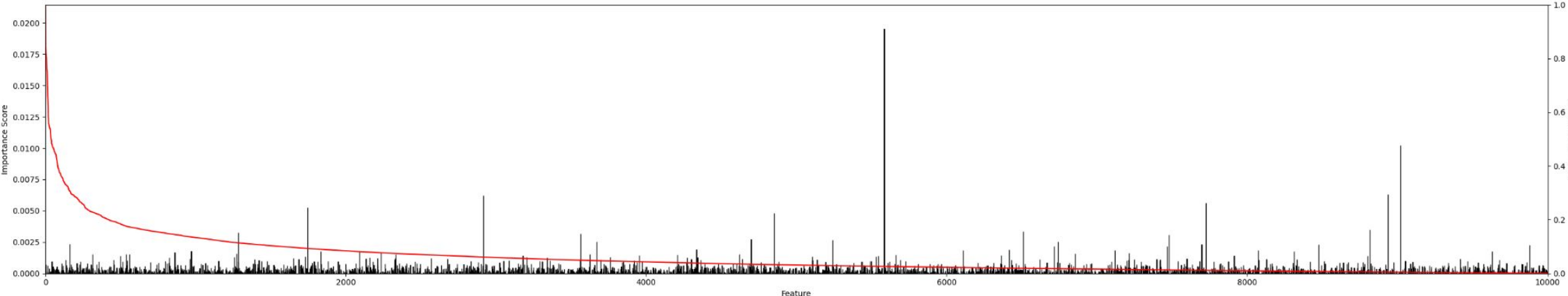
Approach



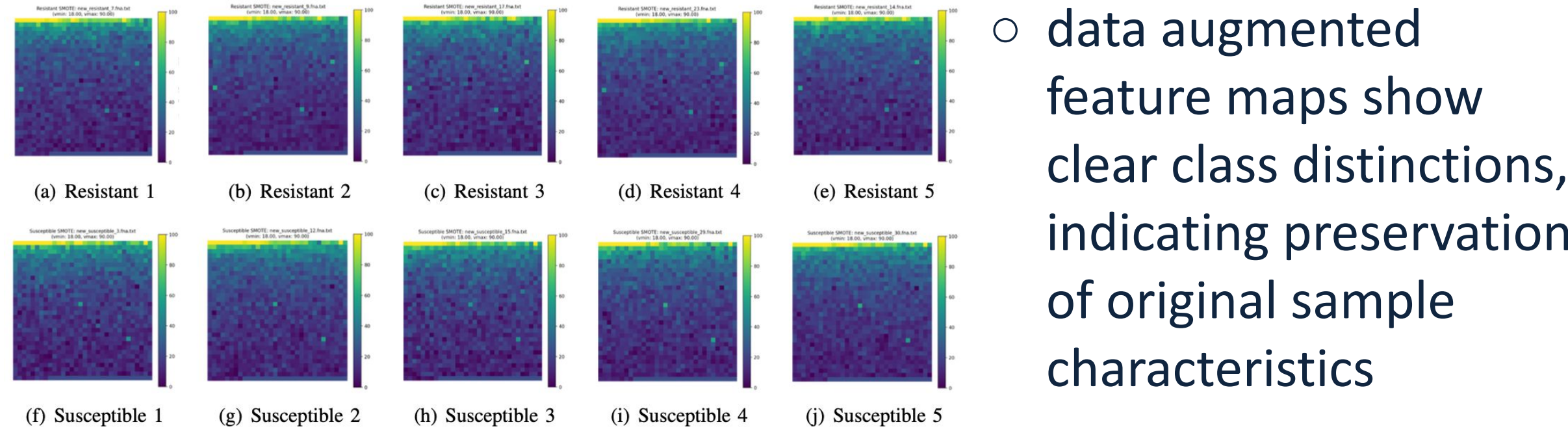
- We utilize n-gram frequency analysis.
- The detailed step is below:
 - Step 1. Download AMR Dataset files from the BV-BRC repository for resistant and susceptible cases.
 - Step 2. Select the number of files each from the resistant and susceptible cases in order of file size out of over 2,000 files.
 - Step 3. Make intermediate files by filtering out meta information.
 - Step 4. Generate k-mers (i.e., 7 or 10-mers) from each file.
 - Step 5. Calculate the frequency of each k-mers across all files and select the top number features out of 1 million features.
 - Step 6. Generate CSV files containing the top number features with labels assigned as 0 for resistant and 1 for susceptible.
 - Step 7. Apply data augmentation method to test its effect on model performance
 - Step 8. Use ensemble ML models to analyze the data.

Visualization

- Frequency vs. Feature Importance
 - Suggests that Frequency doesn't imply Feature Importance



- Data Visualization



- data augmented feature maps show clear class distinctions, indicating preservation of original sample characteristics

Analysis

- This table shows performance differences of ML models and tuning methods across genome-antibiotic combinations.

Genome	Antibiotic	Model	Dataset	Hyperparameter Tuning Method			VanOeffelen <i>et al.</i> [30]
				Bayesian	Randomized	Grid	
E. faecium	vancomycin (VAN)	RF	Original 200	0.8676	0.8420	0.8467	~92% on XGB
		XGB	Original 200	0.9231	0.8472	0.8610	
		ET	Original 200	0.9042	0.8234	0.8267	
E. coli	ampicillin (AMP)	RF	Original 200	0.8412	0.8196	0.8280	~88% on XGB
		XGB	Original 200	0.8380	0.8963	0.8438	
		ET	Original 200	0.8217	0.8340	0.8323	
A. baumannii	amikacin (AMK)	RF	Original 200	0.8150	0.8862	0.8948	~86% on XGB
		XGB	Original 200	0.8107	0.8276	0.8794	
		ET	Original 200	0.9042	0.9019	0.9100	
E. coli	gentamicin (GEN)	RF	Original 200	0.8680	0.9730	0.9778	~75% on XGB
		XGB	Original 200	0.8433	0.8753	0.8602	
		ET	Original 200	0.8217	0.9828	0.9756	
E. coli	trimethoprim/sulfamethoxazole (SXT)	RF	Original 200	0.7296	0.7323	0.7413	~70% on XGB
		XGB	Original 200	0.7432	0.7350	0.7442	
		ET	Original 200	0.7500	0.7395	0.7401	
E. faecium	ampicillin (AMP)	RF	Original 200	0.9744	1.0000	1.0000	~93% on XGB
		XGB	Original 200	0.9745	0.9773	0.9773	
		ET	Original 200	0.9729	0.9829	0.9829	
S. enterica	ampicillin (AMP)	RF	Original 200	0.8756	0.8736	0.8776	~95% on XGB
		XGB	Original 200	0.8552	0.8439	0.8478	
		ET	Original 200	0.8598	0.8611	0.8625	
S. enterica	chloramphenicol (CHL)	RF	Original 200	0.8298	0.8323	0.8369	~95% on XGB
		XGB	Original 200	0.8706	0.8489	0.8718	
		ET	Original 200	0.8394	0.8272	0.8243	

- In six out of eight genome-antibiotic combinations, our pipeline either outperforms or closely matches the accuracy reported by VanOeffelen *et al* [30].
- Using a diverse ensemble of models and tuning methods can maximize performance depending on the dataset.

[30] Margo VanOeffelen, et al., "A genomic data resource for predicting antimicrobial resistance from laboratory-derived antimicrobial susceptibility phenotypes," Briefings in Bioinformatics, vol. 22, no. 6, pp. bbab313, 2021.

- CNN model performance on various genome-antibiotic combinations

Genome	Antibiotic	Dataset	Precision	Recall	F1-score	Accuracy
E. faecium	Vancomycin (VAN)	Original	0.8831	0.8973	0.8875	0.8950
		New 200+100	0.9416	0.9250	0.9330	0.9335
		New 400+100	0.9222	0.9430	0.9319	0.9310
		New 600+100	0.9407	0.9300	0.9352	0.9355
E. coli	ampicillin (AMP)	Original	0.7549	0.9135	0.8263	0.8225
		New 200+100	0.8927	0.9400	0.9157	0.9135
		New 400+100	0.8836	0.9250	0.9029	0.9005
		New 600+100	0.8907	0.9210	0.9052	0.9035
Acinetobacter	amikacin (AMK)	Original	0.8988	0.8432	0.8689	0.8825
		New 200+100	0.9101	0.9370	0.9232	0.9220
		New 400+100	0.9026	0.9440	0.9228	0.9210
		New 600+100	0.8949	0.9420	0.9177	0.9155
E. coli	gentamicin (GEN)	Original	0.8497	0.9081	0.8774	0.8825
		New 200+100	0.9410	0.9470	0.9436	0.9435
		New 400+100	0.9363	0.9500	0.9430	0.9425
		New 600+100	0.9518	0.9400	0.9457	0.9460
E. coli	trimethoprim/sulfamethoxazole (SXT)	Original	0.6564	0.8550	0.7396	0.7000
		New 200+100	0.8622	0.9267	0.8931	0.8890
		New 400+100	0.9235	0.9500	0.9363	0.9354
		New 600+100	0.9597	0.9474	0.9534	0.9537
E. faecium	ampicillin (AMP)	Original	0.9902	1.0000	0.9951	0.9950
		New 200+100	0.9980	0.9946	0.9964	0.9964
		New 400+100	1.0000	0.9956	0.9978	0.9978
		New 600+100	1.0000	0.9968	0.9985	0.9985
Salmonella	ampicillin (AMP)	Original	0.8435	0.8050	0.8215	0.8250
		New 200+100	0.8927	0.9507	0.9203	0.9177
		New 400+100	0.9203	0.9796	0.9489	0.9472
		New 600+100	0.9467	0.9834	0.9645	0.9639
Salmonella	chloramphenicol (CHL)	Original	0.8545	0.9100	0.8813	0.8775
		New 200+100	0.9211	0.9713	0.9455	0.9440
		New 400+100	0.9547	0.9768	0.9656	0.9652
		New 600+100	0.9653	0.9854	0.9753	0.9750

Conclusion

- This study shows that AMR prediction can be improved using cross-domain methods that exploit shared dataset features
- To address the small dataset limitation, data augmentation methods were applied. ML showed competitive F1-scores and CNN demonstrated improved performance.
- These findings suggest that data augmentation methods hold potential for overcoming data scarcity in AMR prediction tasks

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